WEEKLY LESSON PLAN · 2026-27

# Manufacturing Engineering Technology I

TEKS §127.813 (working draft) · CTE Level 2 · Grade 10 · Mr. Gavaldon · Da Vinci School
robertgavaldon@burnhamwood.org

SEMESTER 1
**Week 2**

## MONDAY Day 6 - Measurement & Tools

Aug 17

| | |
|---|---|
| OBJECTIVE | Students will read a metric vernier caliper to a least count of 0.02 mm, distinguish accuracy, precision, and resolution using trial data (mean and range), and apply a bilateral tolerance to accept or reject a part. |
| TEKS | §127.813 (draft, verify strand) measurement, tools & precision in manufacturing |
| ELPS / EB SUPPORT | ELPS/EB supports at-level: pre-teach accuracy/precision/tolerance with a picture card (PV, DP); model the vernier reading step-by-step and repeat slowly (RS, VC); hands-on measuring stations with a partner and native-language number talk (PN, CD); wait time on the classify-the-data exit question (WT). |
| AGENDA · 45 MIN | **6′** Bell-work: estimate this block's thickness by eye, then we will measure it<br>**10′** Precision vs. accuracy vs. resolution; reading a vernier caliper to 0.02 mm<br>**8′** Worked examples: vernier reading + tolerance limits (GO / NO-GO)<br>**16′** Measuring lab: 3 objects x 3 trials, compute mean and range, check tolerance<br>**5′** Exit ticket: classify a data set as accurate and/or precise |
| SLIDES / ACTIVITY | Measure Like a Machinist: Precision, Tolerance & the Vernier Caliper |
| NOTEBOOK | Record your station measurements. Define precision vs. tolerance in your own words. |
| TURN IN | Precision, Tolerance & the Vernier Caliper - Lab + Problem Set |

## TUESDAY Day 7 - Manufacturing Math

Aug 18

| | |
|---|---|
| OBJECTIVE | Students will convert between inches (fractional and decimal) and millimeters using the exact factor 1 in = 25.4 mm, compute tolerance limits and a worst-case tolerance stack-up, and evaluate a clearance fit - showing all work to the correct resolution. |
| TEKS | §127.813 (draft, verify strand) applied mathematics & measurement in manufacturing |
| ELPS / EB SUPPORT | ELPS/EB supports at-level: keep a conversion reference card (1 in = 25.4 mm) and worked-example frames (CD, RS); pre-teach nominal / tolerance / limits / stack-up (PV); dual-language number talk with a partner (PN); the mm = 25.4 x in graph is a visual anchor for the relationship (VC, DP); extra time on the stack-up and fit problems (ET). |
| AGENDA · 45 MIN | **6′** Bell-work: convert 2.5 in to mm (show the setup)<br>**12′** Conversions (1 in = 25.4 mm), fraction<->decimal, tolerance limits and stack-up<br>**8′** Worked examples: convert + tolerance range + clearance fit<br>**14′** Practice set: real part dimensions (show all work)<br>**5′** Go over answers + exit ticket |
| SLIDES / ACTIVITY | The Math Behind the Part: Conversion, Tolerance & Fit |
| NOTEBOOK | Keep the conversion table. Work 2 example problems with all steps shown. |
| TURN IN | Manufacturing Math - Conversion, Tolerance & Fit Problem Set |

## WEDNESDAY Day 8 - Parametric Design in Fusion

Aug 19

| | |
|---|---|
| OBJECTIVE | Students will define user parameters and driving relations (expressions) in Fusion 360, apply geometric and dimensional constraints to fully constrain a sketch, and explain how design intent lets one parameter change regenerate a manufacturable part correctly. |
| TEKS | `§127.813 (draft, verify strand) parametric CAD for manufacturable parts` |
| ELPS / EB SUPPORT | ELPS/EB supports at-level: build steps mirrored on-screen and posted as Classroom screenshots (VC, DP); pre-teach parameter / expression / constraint / fully-constrained (PV); rephrase and slow each build step (RS); finishers pair with peers who are stuck (PN); extra time to reach a fully-constrained sketch (ET). |
| AGENDA · 45 MIN | **6′** Bell-work: change one number and the whole part updates - why?<br>**6′** Watch: Fusion first look - dimensions drive the shape<br>**10′** Parameters, expressions/relations, constraints and fully-constrained sketches<br>**18′** Guided build: parametric plate driven by width / height / thickness / edge<br>**5′** Exit: list your 3 parameters and what each controls; save |
| SLIDES / ACTIVITY | Parametric Design: Parameters, Relations & Constraints |
| NOTEBOOK | Write the 3 parameters you set (name = value) and what each controls on the part. |
| TURN IN | Parametric Design - Parameters, Relations & Constraints |

## THURSDAY Day 9 - Start the Parametric Project (Mounting Plate)

Aug 20

| | |
|---|---|
| OBJECTIVE | Students will write a formal problem statement for a mounting plate, select a clearance hole from a fastener chart, compute plate size, hole coordinates, and an edge-distance check from a component's bolt pattern, and begin modeling the plate parametrically in Fusion 360. |
| TEKS | `§127.813 (draft, verify strand) apply parametric CAD to a defined part; documentation` |
| ELPS / EB SUPPORT | ELPS/EB supports at-level: the part-planning sheet organizes the math before CAD (CD, DP); a reference model and a dimensioned sketch are shown on screen (VC); pre-teach clearance hole / edge distance / datum / fastener designation (PV); partner talk in native language while planning (PN); extra time to finish the sketch (ET). |
| AGENDA · 45 MIN | **5′** Bell-work: name 2 components a plate might hold, and the fastener size<br>**12′** Project brief: problem statement, clearance holes, edge distance, datums, tolerances<br>**8′** Worked example: size a plate from a component's bolt pattern<br>**15′** Plan YOUR dimensions on the sheet, then begin modeling in Fusion<br>**5′** Save (yourname-plate) + exit |
| SLIDES / ACTIVITY | Project: Mounting Plate - From Brief to Parametric Model |
| NOTEBOOK | Record your plate dimensions and hole sizes. Sketch the plate with dimensions labeled. |
| TURN IN | Part-Planning Sheet - Mounting Plate (with calculations) |

**FRIDAY** ## Day 10 - Continue Project + Certification Kickoff

Aug 21

**OBJECTIVE**

Students will map the Autodesk Certified User (Fusion) exam domains to features of their mounting plate, run a parametric robustness test (change a parameter, predict and verify the update), and complete an evidence-based peer review of a partner's model against a rubric.

**TEKS**

§127.813 (draft, verify strand) industry certification pathway; collaboration

**ELPS / EB SUPPORT**

ELPS/EB supports at-level: the certification path is shown as a simple roadmap graphic (VC, DP); pre-teach certification / domain / rubric / performance-based (PV); peer-review pairs an EB student with a helper and native-language discussion is welcome (PN); rephrase and slow the test-strategy tips (RS); wait time on the self-check (WT).

**AGENDA · 45 MIN**

- **5′** Bell-work: what does an industry certification get you?
- **10′** Autodesk Certified User: logistics + the 5 objective domains
- **8′** Worked: parametric robustness test + how a failure shows
- **17′** Continue modeling your plate; run the robustness test; peer-review a partner
- **5′** Self-check + exit

**SLIDES / ACTIVITY**

Certification Kickoff + Parametric Robustness & Peer Review

**NOTEBOOK**

List the certification domains. Note where you are on your plate and one thing to finish Monday.

**TURN IN**

Certification Domains, Robustness Test & Peer Review